Part 1 Introduction to contract testing

In software testing, various testing types provide different values to ensure that when your application is deployed to your users, it meets their expectations, and risks are diminished as much as possible. You're likely familiar with testing types such as unit testing, integration testing, functional testing, and end-to-end testing, so you might wonder why we're introducing contract testing with all of these other testing types out there.

As more teams adopt a microservices architecture, where software projects are broken down into smaller modules or components and developed independently by different teams, the need to test and deploy the changes independently will increase. The microservices architecture poses new challenges to how teams traditionally approach testing activities. Multiple teams need to deploy their service to a dedicated test environment. Imagine if you have a lot of them! How will you test the services efficiently? This is where contract testing shines through.

This part of the book delves further into the concept of contract testing. Chapter 1 provides a high-level overview of contract testing. Chapter 2 discusses how contract testing fits into wider testing concepts and whether it replaces other testing activities. You learn about the numerous benefits and challenges of contract testing. You also learn how to introduce contract testing to your teams. Chapter 3 discusses the technical side of contract testing. You learn what is meant by *consumer*, *provider*, *contract*, and *broker* in this context, as well as get an overview of the contract testing life cycle. You gain an understanding of the various contract testing tools available, the difference between consumer-driven contract testing (CDCT) and provider-driven contract testing (PDCT), and the different communication types that contract testing supports. By the end of this part of the book, you'll be equipped with the core knowledge needed to start implementing contract testing into your projects.